



STEM SMART Physics 18 - Proportionality >

Factor and Percentage Changes 1

Essential Pre-Uni Physics A9.1

Subject & topics: Physics | Skills | Relationships **Stage & difficulty:** GCSE C2, A Level P1

Give the factor by which the quantity changes to two significant figures. So if it doubles, your answer is 2.0, and if it halves your answer is 0.50.

By what factor does $E = mv^2/2$ change if v doubles?

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Factor and Percentage Changes 4

Essential Pre-Uni Physics A9.4

Subject & topics: Physics | Skills | Relationships **Stage & difficulty:** GCSE C3, A Level C1

Give the factor by which the quantity changes to two significant figures. So if it doubles, your answer is 2.0, and if it halves your answer is 0.50.

By what factor does d need to change in $I = P/(4\pi d^2)$ if P were multiplied by 5.2 and I were to not change?

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Factor and Percentage Changes 10

Essential Pre-Uni Physics A9.10

Subject & topics: Physics | Skills | Relationships Stage & difficulty: GCSE C3, A Level C1

Give the percentage change (to two significant figures) in the quantity. Use "+" or "−" to indicate increase or decrease (so a 3.0 % decrease would be given as −3.0 %).

Part A

Percentage change in E

In $E = VIt$, what is the percentage change in E if V increases by 1.0 %, I decreases by 2.0 % and t increases by 3.0 %?

Part B

Percentage change in R

In $R = \rho L/A$, what is the percentage change in R if L increases by 7.0 % and A increases by 3.0 %?

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Proportionality 1

Essential Pre-Uni Physics A10.1

Subject & topics: Physics | Skills | Relationships **Stage & difficulty:** GCSE C3, A Level P1

If $V \propto I$ and $I = 0.35\text{ A}$ when $V = 9.6\text{ V}$, what will V be when $I = 0.90\text{ A}$?

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Proportionality 3

Essential Pre-Uni Physics A10.3

Subject & topics: Physics | Skills | Relationships **Stage & difficulty:** GCSE C3, A Level P1

If $g \propto 1/r^2$ and $g = 9.8 \text{ N kg}^{-1}$ when $r = 6400 \text{ km}$, what will g be when $r = 15000 \text{ km}$?

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Proportionality 4

Essential Pre-Uni Physics A10.4

Subject & topics: Physics | Skills | Relationships **Stage & difficulty:** GCSE C3, A Level P1

If $E \propto x^2$ and $E = 2.5 \text{ J}$ when $x = 1.5 \text{ cm}$, what will x be when $E = 6.0 \text{ J}$?

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Proportionality 4

Essential GCSE Physics 7.4

Subject & topics: Physics | Skills | Relationships Stage & difficulty: Year 9 C2, GCSE C1

The number of widgets made in a factory each week is proportional to the number of workers and the number of hours each worker works. When the factory employed 25 staff, each working 35 hours / week, 65 400 widgets were made each week.

Part A

40 staff for 30 hours/week

How many widgets would be made each week if 40 staff worked for 30 hours per week?

Part B

130000 widgets

If we need 130 000 widgets made each week, and the staff will work 42 hours / week, how many workers are needed?

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Introducing the Doppler Effect

Subject & topics: Physics | Waves & Particles | Wave Motion

Stage & difficulty: GCSE C3, A Level C1

When we listen to a sound from a speaker moving towards or away from us, the frequency we hear is not the same as if the speaker were still. This is called the **Doppler effect**.

In this question you will derive the equations describing the frequency we hear when a speaker is moving away from us or towards us.

A speaker is making a sound with frequency f and time period $T = \frac{1}{f}$. It is moving away from us at a constant speed v .

At time $t = 0$ the speaker is a distance s away from us. It emits a peak.

At time $t = T$ it emits its next peak.

In this question, we use c to represent the speed of sound.

Part A

Receiving the first peak

At what time will we hear the first peak?

The following symbols may be useful: c , s

Part B

Sending the second peak

How far away from us is the speaker when it sends the second peak?

The following symbols may be useful: T , s , v

When will we hear the second peak?

The following symbols may be useful: T , c , s , v

Part C

Time delay

The time which passes between our hearing the first peak and hearing the second peak is the time period of the wave as we hear it T .

What is the time period T of the wave as we hear it?

Give your answer in terms of T , v and c .

The following symbols may be useful: T , c , v

What is the frequency f of the wave as we hear it?

Give your answer in terms of f , v and c .

The following symbols may be useful: c , f , v

Part D

Speaker moving towards us

So far, we have thought about a speaker moving away from us. Now we think about a speaker moving towards us. Suppose that when $t = 0$, the speaker is at a distance s from us, and it is moving towards us at speed v . The speaker emits a peak when $t = 0$, and then emits its next peak when $t = T$. The second peak is emitted while the speaker is still moving towards us.

The first peak is received when $t = s/c$. When is the second peak received?

The following symbols may be useful: T , c , s , v

What is the frequency we hear f in this case?

The following symbols may be useful: c , f , v

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